ECON 5021 Unit 2 Study Guide

Linear Regression Model and OLS (Lectures 8–12): the proof bank

Built from the Canvas short-answer questions (SAQs) and the lecture notes (Oct 14, 2026 version)

How to use this guide

The proof section of the exam is 3–5 proofs taken from the SAQs, sometimes with the parts reordered or combined. This guide gives you:

1. **§1 Toolkit**: the 12 algebra facts that every proof uses. Learn these first.
2. **§2 Proof bank (P1–P15)**: every SAQ proof, written out in full. The SAQ answers that only said *"see the lecture notes"* or *"work this out on your own"* are filled in here from the notes. Each proof starts with a **skeleton card** (the chain of steps to memorize) and ends with any common traps.
3. **§3 One-page formula sheet** and **§4 Skeleton drill**: for the night before.
4. **§5 Mock proof exam**: 5 permuted prompts. Write them out cold, then check against the bank.

Method: read a skeleton card, close the guide, write the full proof on blank paper, then compare. Any proof you can't reproduce twice in a row goes back in the pile.

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1 Toolkit: the moves every proof uses

T1. Transpose of a product: $(AB)' = B'A'$. Also $(A^{-1})' = (A')^{-1}$, so $(X'X)^{-1}$ is symmetric because $X'X$ is.

T2. Inverse of a scaled matrix: $(cA)^{-1} = c^{-1}A^{-1}$. This is how you pull n out: $(X'X)^{-1} = \left(n \cdot \frac{X'X}{n}\right)^{-1} = \frac{1}{n}\left(\frac{X'X}{n}\right)^{-1}$.

T3. 2×2 inverse: $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \dfrac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

T4. Constants come out of (conditional) expectations: $\mathrm{E}(a + Bz) = a + B\,\mathrm{E}(z)$. Conditioning on X makes any function of X (such as $A = (X'X)^{-1}X'$) a constant: $\mathrm{E}(A\boldsymbol{\varepsilon} \mid X) = A\,\mathrm{E}(\boldsymbol{\varepsilon} \mid X)$.

T5. Law of iterated expectations (LIE): $\mathrm{E}(z) = \mathrm{E}_X\big[\mathrm{E}(z \mid X)\big]$. This is how you go from a *conditional* result to an *unconditional* one.

T6. Variance of a linear transform: $\mathrm{Var}(a + Az \mid X) = A\,\mathrm{Var}(z \mid X)\,A'$.

T7. Variance as an outer product: $\mathrm{Var}(z) = \mathrm{E}(zz') - \mathrm{E}(z)\mathrm{E}(z)'$. When $\mathrm{E}(z) = 0$, this becomes $\mathrm{Var}(z) = \mathrm{E}(zz')$.

T8. Projection and residual maker: $P_X = X(X'X)^{-1}X'$ and $M_X = I - P_X$. Both are *symmetric* ($P' = P$) and *idempotent* ($PP = P$). Also $P_X X = X$, $M_X X = 0$, $P_X Y = Xb = \hat{Y}$, and $M_X Y = e$.

T9. Trace rules: a scalar equals its own trace. $\mathrm{tr}(AB) = \mathrm{tr}(BA)$ (you can cycle the factors). $\mathrm{tr}(A \pm B) = \mathrm{tr}A \pm \mathrm{tr}B$, $\mathrm{tr}(cA) = c\,\mathrm{tr}A$, $\mathrm{tr}(I_n) = n$, and $\mathrm{E}[\mathrm{tr}(\cdot)] = \mathrm{tr}[\mathrm{E}(\cdot)]$.

T10. Sums as matrix products: $X'X = \sum_i x_i x_i'$, $X'Y = \sum_i x_i y_i$, and $X'\boldsymbol{\varepsilon} = \sum_i x_i \varepsilon_i$. For a column of ones ι: $\iota'\iota = n$, $\iota' x = \sum_i x_i = n\bar{x}$.

T11. Sample covariance and variance shortcuts: $\frac{1}{n}\sum(x_i - \bar{x})(y_i - \bar{y}) = \overline{xy} - \bar{x}\bar{y}$ and $\sum(x_i - \bar{x})^2 = \sum x_i^2 - n\bar{x}^2$. To prove the second: expand, and use $\sum x_i = n\bar{x}$.

T12. Completing the square / cross-term trick (used in P1 and P2): write the target as *(the true thing) + (the error)*, expand the square, then show the cross term has expectation zero.

The five LRM assumptions (you are often asked which one a step uses): **A1** linearity, $Y = X\beta + \boldsymbol{\varepsilon}$. **A2** full rank, so $(X'X)^{-1}$ exists. **A3** mean independence, $\mathrm{E}(\boldsymbol{\varepsilon} \mid X) = 0$. **A4** homoskedasticity and no autocorrelation, $\mathrm{Var}(\boldsymbol{\varepsilon} \mid X) = \sigma^2 I_n$. **A5** normality, $\boldsymbol{\varepsilon} \mid X \sim N(0, \sigma^2 I_n)$. Unbiasedness needs A1–A3. $\mathrm{Var}(b \mid X) = \sigma^2(X'X)^{-1}$ also needs A4. Exact N/t inference also needs A5.

2 Proof bank

Lecture 8: Linear Regression Model

P1. The mean minimizes expected squared error

L8 Q1

Show: $\arg\min_a \mathrm{E}[(y - a)^2] = \mu = \mathrm{E}(y)$.

Skeleton

$y = \mu + u$ with $\mathrm{E}u = 0 \;\to\; \mathrm{E}[(u + \mu - a)^2] \;\to$ expand $\to$ the cross term $2(\mu - a)\mathrm{E}u = 0 \;\to$ $\mathrm{E}u^2 + (\mu - a)^2 \;\to$ set $a = \mu$.

Define $u = y - \mu$. Then $\mathrm{E}(u) = \mathrm{E}(y) - \mu = 0$. Substitute:

$$\mathrm{E}[(y-a)^2] = \mathrm{E}[(u+\mu-a)^2] = \mathrm{E}\big[u^2 + 2u(\mu-a) + (\mu-a)^2\big] = \mathrm{E}(u^2) + 2(\mu-a)\underbrace{\mathrm{E}(u)}_{0} + (\mu-a)^2.$$

So $\mathrm{E}[(y-a)^2] = \mathrm{E}(u^2) + (\mu-a)^2$. The first term does not depend on a. The second term is ≥ 0, and it equals 0 exactly when $a = \mu$. ■

Calculus check: $\frac{d}{da}\mathrm{E}[(y-a)^2] = -2\mathrm{E}(y-a) = 0 \Rightarrow a = \mathrm{E}y$. Use the decomposition argument on the exam, because it is the one the solution key uses.

P2. The CEF is the best predictor

L8 Q2(a)–(c)

Setup: $m(x) = \mathrm{E}(y \mid x)$ and $u = y - m(x)$.

Skeleton

(a) $\mathrm{E}(u|x) = \mathrm{E}(y|x) - m(x) = 0$. (b) $\mathrm{E}(h(x)u|x) = h(x)\mathrm{E}(u|x) = 0$, then use LIE. (c) $y = m + u$ $\to$ expand $\mathrm{E}[(u + m - g)^2] \to$ the cross term is $\mathrm{E}[u\,h(x)]$ with $h = m - g$, which is 0 by (b) $\to$ $\mathrm{E}u^2 + \mathrm{E}(m-g)^2 \to$ set $g = m$.

(a) $\mathrm{E}(u \mid x) = \mathrm{E}(y - m(x) \mid x) = \mathrm{E}(y \mid x) - m(x) = m(x) - m(x) = 0$. ($m(x)$ is fixed once we condition on x.)

(b) Condition on x: $\mathrm{E}(h(x)u \mid x) = h(x)\mathrm{E}(u \mid x) = 0$. By the LIE, $\mathrm{E}(h(x)u) = \mathrm{E}_x[\mathrm{E}(h(x)u \mid x)] = \mathrm{E}_x(0) = 0$.

(c) Substitute $y = m(x) + u$:

$$\mathrm{E}[(y - g(x))^2] = \mathrm{E}[(u + m(x) - g(x))^2] = \mathrm{E}(u^2) + 2\mathrm{E}\big[u\,(m(x) - g(x))\big] + \mathrm{E}\big[(m(x) - g(x))^2\big].$$

Let $h(x) = m(x) - g(x)$. The middle term is $2\mathrm{E}[h(x)u] = 0$ by (b). So $\mathrm{E}[(y - g(x))^2] = \mathrm{E}(u^2) + \mathrm{E}[(m(x) - g(x))^2]$. The first term does not depend on g. The second term is minimized, at zero, by $g(x) = m(x) = \mathrm{E}(y \mid x)$. ■

Watch out

You must *name* $h(x) = m(x) - g(x)$ so you can cite part (b). This is the step the grader looks for. Say "by the LIE" explicitly in (b).

P3. Algebraic properties of OLS residuals

L8 Q3(a)–(c)

Given the FOCs $-2\sum_i x_{ik}(y_i - x_i'b) = 0$ for every k, and $e_i = y_i - x_i'b$.

Skeleton

FOCs $\Leftrightarrow \sum_i x_{ik}e_i = 0$ for all k (a) take $k = 1$ with $x_{i1} = 1$, so $\sum e_i = 0$ (b) $\bar{e} = \bar{y} - \bar{x}'b = 0$ (c) expand the covariance into $\frac{1}{n}\sum x_{ik}e_i - \bar{x}_k\bar{e} = 0 - 0$.

Divide each FOC by -2 and substitute e_i: $\sum_{i=1}^n x_{ik}e_i = 0$ for $k = 1, \ldots, K$. (In matrix form, $X'e = 0$.)

(a) With $x_{i1} = 1$, the $k = 1$ condition reads $\sum_i 1 \cdot e_i = 0$, so $\sum_i e_i = 0$ (and $\bar{e} = 0$).

(b) Average the residuals: $\bar{e} = \frac{1}{n}\sum_i (y_i - x_i'b) = \frac{1}{n}\sum_i y_i - \left(\frac{1}{n}\sum_i x_i\right)'b = \bar{y} - \bar{x}'b$. From (a), $\bar{e} = 0$, so $\bar{y} = \bar{x}'b$. In words, the fitted line passes through the point of means.

(c) Expand:

$$\widehat{\mathrm{Cov}}(x_k, e) = \frac{1}{n}\sum_i (x_{ik} - \bar{x}_k)(e_i - \bar{e}) = \frac{1}{n}\sum_i x_{ik}e_i - \bar{x}_k\bar{e} - \bar{e}\bar{x}_k + \bar{x}_k\bar{e} = \underbrace{\frac{1}{n}\sum_i x_{ik}e_i}_{=0\ \text{(FOC)}} - \bar{x}_k \underbrace{\bar{e}}_{=0\ \text{(a)}} = 0.$$

■

Watch out

Part (c) needs *both* the k-th FOC and $\bar{e} = 0$. The second fact comes from having a constant in the model. Without a constant, $X'e = 0$ still holds but $\bar{e}$ need not be 0, so say where each zero comes from.

P4. Bivariate OLS in matrix form; slope = $\widehat{\mathrm{Cov}}/\widehat{\mathrm{Var}}$ L8 Q4(a)–(b)

Setup: $\hat{w} = b_1 + b_2 e$. Here e is *education*, not the residual. $X = [\iota \;\; e]$ and $Y = w$.

Skeleton

Multiply out $X'X$ and $X'Y$ → factor out n → $(nA)^{-1}(nc) = A^{-1}c$ → 2×2 inverse with determinant $\overline{ee} - \bar{e}^2$ → the 2nd row gives $b_2 = (\overline{ew} - \bar{e}\bar{w})/(\overline{ee} - \bar{e}^2)$.

(a) Write out the products element by element:

$$X'X = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ e_1 & e_2 & \cdots & e_n \end{bmatrix} \begin{bmatrix} 1 & e_1 \\ \vdots & \vdots \\ 1 & e_n \end{bmatrix} = \begin{bmatrix} \sum_i 1 & \sum_i e_i \\ \sum_i e_i & \sum_i e_i^2 \end{bmatrix} = n \begin{bmatrix} 1 & \bar{e} \\ \bar{e} & \overline{ee} \end{bmatrix},$$

$$X'Y = \begin{bmatrix} 1 & \cdots & 1 \\ e_1 & \cdots & e_n \end{bmatrix} \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix} = \begin{bmatrix} \sum_i w_i \\ \sum_i e_i w_i \end{bmatrix} = n \begin{bmatrix} \bar{w} \\ \overline{ew} \end{bmatrix}.$$

Then (T2) gives $b = (X'X)^{-1}X'Y = \frac{1}{n}\begin{bmatrix} 1 & \bar{e} \\ \bar{e} & \overline{ee} \end{bmatrix}^{-1} n \begin{bmatrix} \bar{w} \\ \overline{ew} \end{bmatrix} = \begin{bmatrix} 1 & \bar{e} \\ \bar{e} & \overline{ee} \end{bmatrix}^{-1} \begin{bmatrix} \bar{w} \\ \overline{ew} \end{bmatrix}$.

(b) Apply the 2×2 inverse (T3). The determinant is $\overline{ee} - \bar{e}^2$:

$$b = \frac{1}{\overline{ee} - \bar{e}^2} \begin{bmatrix} \overline{ee} & -\bar{e} \\ -\bar{e} & 1 \end{bmatrix} \begin{bmatrix} \bar{w} \\ \overline{ew} \end{bmatrix} = \frac{1}{\overline{ee} - \bar{e}^2} \begin{bmatrix} \overline{ee}\,\bar{w} - \bar{e}\,\overline{ew} \\ -\bar{e}\bar{w} + \overline{ew} \end{bmatrix}.$$

The second element is $b_2 = \dfrac{\overline{ew} - \bar{e}\bar{w}}{\overline{ee} - \bar{e}^2} = \dfrac{\widehat{\mathrm{Cov}}(e, w)}{\widehat{\mathrm{Var}}(e)}$. ■

Bonus check: the first element simplifies to $b_1 = \bar{w} - b_2\bar{e}$. This matches $\bar{y} = \bar{x}'b$ from P3.

Watch out

Don't confuse education e_i with the residual. If you want, rename it ed_i on your paper and say so.

Lecture 9: Properties of OLS

P5. OLS is unbiased (conditionally, then unconditionally) L9 Q1(a)–(b)

Skeleton

Substitute $Y = X\beta + \varepsilon \to b = \beta + (X'X)^{-1}X'\varepsilon \to$ take $\mathrm{E}(\cdot|X)$ and pull out the functions of X $\to \mathrm{E}(\varepsilon|X) = 0$ (A3) $\to \mathrm{E}(b|X) = \beta \to$ LIE gives $\mathrm{E}(b) = \mathrm{E}_X(\beta) = \beta$.

(a)

$$\mathrm{E}(b \mid X) = \mathrm{E}\big[(X'X)^{-1}X'(X\beta + \varepsilon) \mid X\big] = \underbrace{(X'X)^{-1}X'X}_{I_K}\beta + (X'X)^{-1}X' \underbrace{\mathrm{E}(\varepsilon \mid X)}_{0 \text{ by A3}} = \beta.$$

We can pull $(X'X)^{-1}X'$ out of the expectation because it is a function of X, so it is fixed once we condition on X (T4). β is a constant.

(b) By the LIE (T5), $\mathrm{E}(b) = \mathrm{E}_X[\mathrm{E}(b \mid X)] = \mathrm{E}_X(\beta) = \beta$. ■

P6. Sampling error, the variance sandwich, and $\sigma^2(X'X)^{-1}$ L9 Q2(a)–(d)

Skeleton

(a) $b = \beta + A\varepsilon$ with $A = (X'X)^{-1}X'$. (b) $b - \mathrm{E}(b|X) = A\varepsilon$, so $\mathrm{Var} = \mathrm{E}[A\varepsilon\varepsilon'A'|X] = A\,\mathrm{E}(\varepsilon\varepsilon'|X)A'$. (c) $\mathrm{Var} = \mathrm{E}(\varepsilon\varepsilon') - \mathrm{E}\varepsilon\mathrm{E}\varepsilon'$ and the 2nd term is 0. (d) diagonal $= \sigma^2$, off-diagonal $= 0 \to \sigma^2 I \to \sigma^2(X'X)^{-1}X'X(X'X)^{-1} = \sigma^2(X'X)^{-1}$.

(a) $b = (X'X)^{-1}X'(X\beta + \varepsilon) = (X'X)^{-1}X'X\beta + (X'X)^{-1}X'\varepsilon = \beta + (X'X)^{-1}X'\varepsilon$.

(b) Since $\mathrm{E}(b \mid X) = \beta$, we have $b - \mathrm{E}(b \mid X) = (X'X)^{-1}X'\varepsilon$. Its transpose is $\varepsilon'X(X'X)^{-1}$ (T1, using the symmetry of $(X'X)^{-1}$). So

$$\mathrm{Var}(b \mid X) = \mathrm{E}\big[(X'X)^{-1}X'\varepsilon\varepsilon'X(X'X)^{-1} \mid X\big] = (X'X)^{-1}X'\,\mathrm{E}(\varepsilon\varepsilon' \mid X)\,X(X'X)^{-1}.$$

(c) $\mathrm{Var}(\varepsilon \mid X) = \mathrm{E}(\varepsilon\varepsilon' \mid X) - \mathrm{E}(\varepsilon \mid X)\mathrm{E}(\varepsilon \mid X)'$. Under A3 the second term is $0 \cdot 0' = 0$, so $\mathrm{Var}(\varepsilon \mid X) = \mathrm{E}(\varepsilon\varepsilon' \mid X)$.

(d) $\mathrm{Var}(\varepsilon \mid X)$ is $n \times n$. Its diagonal entries are $\mathrm{Var}(\varepsilon_i \mid X)$, and its off-diagonal entries are $\mathrm{Cov}(\varepsilon_i, \varepsilon_j \mid X)$. *Homoskedasticity* means every diagonal entry equals the same σ^2, which does not depend on X. *No autocorrelation* means every off-diagonal entry is 0. So

$$\mathrm{Var}(\varepsilon \mid X) = \begin{bmatrix} \sigma^2 & 0 & \cdots & 0 \\ 0 & \sigma^2 & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma^2 \end{bmatrix} = \sigma^2 I_n.$$

Plug this into (b): $\mathrm{Var}(b \mid X) = (X'X)^{-1}X'(\sigma^2 I_n)X(X'X)^{-1} = \sigma^2(X'X)^{-1}\underbrace{X'X(X'X)^{-1}}_{I_K} = \sigma^2(X'X)^{-1}$.

■

P7. OLS as a sample average: LLN ⇒ consistency, CLT ⇒ normality L9 Q3(a)–(c)

Skeleton

$X'\boldsymbol{\varepsilon} = \sum x_i\varepsilon_i = n\bar{w} \to b = \beta + (X'X)^{-1}n\bar{w} = \beta + (X'X/n)^{-1}\bar{w} \to$ LLN: $\bar{w} \to \mathrm{E}w_i = 0$ and $X'X/n \to \mathrm{E}(x_ix_i')$, so $\operatorname{plim} b = \beta \to$ CLT: $\bar{w}$ is approximately normal, so b is approximately normal even without A5.

(a) $X'\boldsymbol{\varepsilon} = \begin{bmatrix} \sum_i x_{i1}\varepsilon_i \\ \vdots \\ \sum_i x_{iK}\varepsilon_i \end{bmatrix} = \sum_i x_i\varepsilon_i = \sum_i w_i = n\bar{w}$. So $b = \beta + (X'X)^{-1}n\bar{w} = \beta + \left(n^{-1}X'X\right)^{-1}\bar{w} = \beta + \left(\frac{X'X}{n}\right)^{-1}\bar{w}$, using T2.

(b) LLN ⇒ consistency. First, $\mathrm{E}(w_i) = \mathrm{E}(x_i\varepsilon_i) = \mathrm{E}_x[x_i\mathrm{E}(\varepsilon_i \mid x_i)] = 0$ by A3 and the LIE. The LLN says sample averages converge in probability to population means:

- $\operatorname{plim}\bar{w} = \mathrm{E}(w_i) = 0$;
- $\frac{X'X}{n} = \frac{1}{n}\sum_i x_ix_i'$ is also a sample average, so $\operatorname{plim}\frac{X'X}{n} = \mathrm{E}(x_ix_i')$, which is invertible by A2.

Therefore $\operatorname{plim} b = \beta + \mathrm{E}(x_ix_i')^{-1}\cdot 0 = \beta$. **OLS is consistent:** as $n \to \infty$, b concentrates on β. Formally, $\Pr(|b-\beta| \geq \eta) \to 0$ for every $\eta > 0$.

(c) CLT ⇒ approximate normality. $\bar{w}$ is a sample average of mean-zero random vectors, so by the CLT its sampling distribution is approximately normal in large samples. The estimator b is β plus a (nearly constant) matrix $(X'X/n)^{-1}$ times $\bar{w}$. A linear function of an approximately normal vector is approximately normal, so $b \overset{a}{\sim} N\left(\beta, \sigma^2(X'X)^{-1}\right)$ in large samples, *without* assuming A5. This justifies t-tests and confidence intervals when the errors are not normal.

Formal version (optional): $\sqrt{n}\,\bar{w} \xrightarrow{d} N(0, \sigma^2Q)$ with $Q = \mathrm{E}(x_ix_i')$ under A4, so $\sqrt{n}(b-\beta) \xrightarrow{d} N(0, Q^{-1}\sigma^2QQ^{-1}) = N(0, \sigma^2Q^{-1})$.

Related (L9.20): with A5, $b = \beta + A\boldsymbol{\varepsilon}$ is a linear function of a normal vector, so $b \mid X \sim N(\beta, \sigma^2(X'X)^{-1})$ holds *exactly*.

Lecture 10: Interpretation of OLS

P8. Residuals are orthogonal to X: $P_XH = 0$ and regressing e on X gives $a = 0$ L10 Q1(a)–(d)

Skeleton

$P_X = X(X'X)^{-1}X' \to H = Y - Xb \to P_XH = X(X'X)^{-1}X'Y - X(X'X)^{-1}X'Xb = Xb - Xb = 0 \to a = (X'X)^{-1}X'H = b - b = 0$.

(a) $P_X = X(X'X)^{-1}X'$ $(n \times n)$. It projects any n-vector onto the column space of X.

(b) $P_XH = X(X'X)^{-1}X'(Y - Xb) = X\underbrace{(X'X)^{-1}X'Y}_{b} - X\underbrace{(X'X)^{-1}X'X}_{I}b = Xb - Xb = 0.$

(c) In words: the OLS residuals cannot be predicted by *any* linear combination of the columns of X. Everything X can linearly explain about Y is already in $\hat{Y} = Xb$.

(d) Regress H on X: $a = (X'X)^{-1}X'H = (X'X)^{-1}X'Y - (X'X)^{-1}X'Xb = b - b = 0$. ■

(Equivalently, $X'H = X'e = 0$ is just the stacked OLS FOCs from P3.)

P9. Projecting on a constant = averaging; residual maker = demeaning L10 Q2(a)–(c)

Setup: $Y = \tau Z + \kappa X + \varepsilon$, where $Z = \iota$ is an $n \times 1$ vector of ones.

Skeleton

$Z'Z = n \to P_Z = \frac{1}{n}\iota\iota'$, an $n \times n$ matrix with all entries $1/n \to P_Z X = \bar{x}\,\iota \to X^* = M_Z X = X - \bar{x}\iota$, whose i-th entry is $x_i - \bar{x}$.

(a) $Z'Z = \sum_i 1 \cdot 1 = n$, so $P_Z = Z(Z'Z)^{-1}Z' = \frac{1}{n}ZZ'$. Here ZZ' is $(n \times 1)(1 \times n) = n \times n$ with every entry equal to $1 \cdot 1 = 1$. So P_Z **is** $n \times n$ **and every element equals** $1/n$.

(b) $P_Z X$ is $(n \times n)(n \times 1) = n \times 1$. Row i is $\sum_j \frac{1}{n} x_j = \bar{x}$, so $P_Z X = (\bar{x}, \bar{x}, \ldots, \bar{x})'$. **Every element is the sample mean of** X**.**

(c) *(The "work this out on your own" part.)* $M_Z = I - P_Z$, so

$$X^* = M_Z X = X - P_Z X = \begin{bmatrix} x_1 - \bar{x} \\ x_2 - \bar{x} \\ \vdots \\ x_n - \bar{x} \end{bmatrix}.$$

X^* **is** $n \times 1$**, and each element is that observation's price minus the average price**, so X^* is *demeaned* price. The same holds for Y^* $(y_i - \bar{y})$ and ε^* $(\varepsilon_i - \bar{\varepsilon})$. The constant drops out because $M_Z Z = Z - Z = 0$.

Payoff (Frisch–Waugh): OLS on the transformed model gives

$$\hat{\kappa} = (X^{*\prime}X^*)^{-1}X^{*\prime}Y^* = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2} = \frac{\widehat{\mathrm{Cov}}(x, y)}{\widehat{\mathrm{Var}}(x)}.$$

This matches P4: including a constant is the same as demeaning everything.

Lecture 11: Inference with OLS

P10. The standardized estimate: $N(0,1)$, then t_{n-K} L11 Q1

Skeleton

$b_k | X \sim N(\beta_k, \sigma^2[(X'X)^{-1}]_{kk}) \to$ subtract the mean and divide by the SD to get $N(0,1) \to \sigma^2$ is unknown, so use $s^2 = e'e/(n-K) \to$ the result is t_{n-K}.

From $b \mid X \sim N(\beta, \sigma^2(X'X)^{-1})$, the k-th element satisfies $b_k \mid X \sim N\big(\beta_k, \sigma^2[(X'X)^{-1}]_{kk}\big)$. Here $[(X'X)^{-1}]_{kk}$ is the k-th diagonal element. Standardize by subtracting the mean and dividing by the standard deviation:

$$z_k = \frac{b_k - \beta_k}{\sqrt{\sigma^2[(X'X)^{-1}]_{kk}}} \sim N(0,1).$$

σ^2 is unobserved, so estimate it with $s^2 = \frac{1}{n-K}\sum_i e_i^2$ (unbiased, see P11). Replacing σ^2 by s^2 adds estimation noise, and the distribution becomes

$$t_k = \frac{b_k - \beta_k}{\sqrt{s^2[(X'X)^{-1}]_{kk}}} = \frac{b_k - \beta_k}{\widehat{SE}(b_k)} \sim t_{n-K}.$$

Why t (optional depth): $t_k = z_k / \sqrt{\frac{(n-K)s^2/\sigma^2}{n-K}}$. Under A5, $(n-K)s^2/\sigma^2 = \boldsymbol{\varepsilon}' M_X \boldsymbol{\varepsilon}/\sigma^2 \sim \chi^2_{n-K}$, and it is independent of b. A standard normal divided by $\sqrt{\chi^2_{df}/df}$ is t_{df}.

How it's used: 95% CI is $b_k \pm \kappa \widehat{SE}(b_k)$ with $\kappa =$ `qt(0.975, n-K)`. To test $H_0 : \beta_k = q$, use $t = (b_k - q)/\widehat{SE}$ with p-value `2*(1-pt(abs(t), n-K))`.

P11. $e'e = \boldsymbol{\varepsilon}' M_X \boldsymbol{\varepsilon}$ and s^2 is unbiased (trace proof) — L11 Q2(a)–(b)

Skeleton

(a) $e = M\boldsymbol{\varepsilon} \to e'e = \boldsymbol{\varepsilon}' M' M \boldsymbol{\varepsilon} \to M' = M$, and expand $MM = I - 2P + PP = I - P = M$.
(b) scalar = its trace $\to$ cycle the factors $\to$ E and tr commute $\to \sigma^2 \text{tr}(M) \to \text{tr}(M) = n - \text{tr}((X'X)^{-1}X'X) = n - K$.

(a) (Why $e = M_X \boldsymbol{\varepsilon}$: $e = M_X Y = M_X(X\beta + \boldsymbol{\varepsilon}) = M_X X\beta + M_X \boldsymbol{\varepsilon} = M_X \boldsymbol{\varepsilon}$, since $M_X X = X - X(X'X)^{-1}X'X = 0$.) Then $e'e = (M_X\boldsymbol{\varepsilon})'(M_X\boldsymbol{\varepsilon}) = \boldsymbol{\varepsilon}' M_X' M_X \boldsymbol{\varepsilon}$. *Symmetry:* $M_X' = I' - \big(X(X'X)^{-1}X'\big)' = I - X\big((X'X)^{-1}\big)'X' = I - X(X'X)^{-1}X' = M_X$. *Idempotency:*

$$\begin{aligned} M_X' M_X = M_X M_X &= \big(I - X(X'X)^{-1}X'\big)\big(I - X(X'X)^{-1}X'\big) \\ &= I - 2X(X'X)^{-1}X' + X\underbrace{(X'X)^{-1}X'X}_{I}(X'X)^{-1}X' \end{aligned}$$

$$= I - 2X(X'X)^{-1}X' + X(X'X)^{-1}X' = I - X(X'X)^{-1}X' = M_X.$$

So $e'e = \boldsymbol{\varepsilon}' M_X \boldsymbol{\varepsilon}$.

(b) *(Deriving the given expression, in case they ask:)* $e'e$ is a scalar, so $e'e = \text{tr}(\boldsymbol{\varepsilon}' M_X \boldsymbol{\varepsilon}) = \text{tr}(M_X \boldsymbol{\varepsilon}\boldsymbol{\varepsilon}')$ by cycling. Since E and tr commute and M_X is fixed given X: $\text{E}(e'e \mid X) = \text{tr}\big(M_X \text{E}(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid X)\big)$.

Under A4 (homoskedasticity and no autocorrelation), $\text{E}(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}' \mid X) = \text{Var}(\boldsymbol{\varepsilon} \mid X) = \sigma^2 I_n$. The first equality uses A3, as in P6(c). Then

$$\text{E}(e'e \mid X) = \text{tr}(M_X \sigma^2 I_n) = \sigma^2 \text{tr}(M_X).$$

Compute the trace:

$$\text{tr}(M_X) = \text{tr}(I_n) - \text{tr}\big(X(X'X)^{-1}X'\big) = n - \text{tr}\big((X'X)^{-1}X'X\big) = n - \text{tr}(I_K) = n - K.$$

In the middle step, cycle X from the front to the back. Note that I_K is $K \times K$ because $(X'X)^{-1}X'X$ is $K \times K$. So $\text{E}(e'e \mid X) = \sigma^2(n-K)$. Dividing by the constant $n - K$:

$$\text{E}\Big(\frac{e'e}{n-K} \,\Big|\, X\Big) = \sigma^2, \qquad \text{i.e. } s^2 \text{ is (conditionally) unbiased. By the LIE, } \text{E}(s^2) = \sigma^2. \quad \blacksquare$$

Watch out

"No shortcuts" means: write that M_X is symmetric, show $MM = M$ by expanding, and justify each trace step: scalar = trace, cycling, linearity, $\text{tr}(I_n) = n$, $\text{tr}(I_K) = K$. The dimension switch from $n \times n$ to $K \times K$ is the key point, so state it.

P12. Variance of the bivariate slope: $\sigma^2/\sum(x_i-\bar{x})^2$ L11 Q3(a)–(b)

Skeleton

$X'X = \begin{bmatrix} n & \sum x \\ \sum x & \sum x^2 \end{bmatrix}$ → 2 × 2 inverse, determinant $n\sum x^2 - (\sum x)^2$ → the (2,2) entry is $n/\det$ → $\det = n\sum(x_i-\bar{x})^2$ → $\sigma^2/SST_x = \sigma^2/[(n-1)s_x^2]$.

(a) $X'X = \begin{bmatrix} 1 & \cdots & 1 \\ x_1 & \cdots & x_n \end{bmatrix} \begin{bmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} = \begin{bmatrix} n & \sum_i x_i \\ \sum_i x_i & \sum_i x_i^2 \end{bmatrix}$. By T3,

$$\mathrm{Var}(b \mid X) = \sigma^2(X'X)^{-1} = \frac{\sigma^2}{n\sum_i x_i^2 - (\sum_i x_i)^2} \begin{bmatrix} \sum_i x_i^2 & -\sum_i x_i \\ -\sum_i x_i & n \end{bmatrix}.$$

$\mathrm{Var}(b_2 \mid X)$ is the (2,2) element: $\dfrac{\sigma^2\, n}{n\sum_i x_i^2 - (\sum_i x_i)^2}$. Simplify the determinant using $\sum_i x_i = n\bar{x}$:

$$n\sum x_i^2 - (n\bar{x})^2 = n\Big(\sum x_i^2 - n\bar{x}^2\Big) = n\sum_i (x_i-\bar{x})^2.$$

The last step is T11: $\sum(x_i-\bar{x})^2 = \sum x_i^2 - 2\bar{x}\sum x_i + n\bar{x}^2 = \sum x_i^2 - n\bar{x}^2$. Cancel n:

$$\mathrm{Var}(b_2 \mid X) = \frac{\sigma^2}{\sum_{i=1}^n (x_i-\bar{x})^2} = \frac{\sigma^2}{SST_x}. \quad \blacksquare$$

(b) Three factors. Write $SST_x = (n-1)s_x^2$, so $\mathrm{Var}(b_2 \mid X) = \dfrac{\sigma^2}{(n-1)s_x^2}$.

1. σ^2, the **disturbance variance**: more noise in the unobservables gives a less precise b_2.
2. n, the **sample size**: more observations give a smaller variance, holding s_x^2 fixed.
3. s_x^2, the **variation in** x: more spread in the regressor gives a more precise slope, holding n fixed.

P13. Multiple regression: $\mathrm{Var}(b_k) = \sigma^2/[(1-R_k^2)SST_{x_k}]$ L11 Q4(a)–(b)

Skeleton

Partition $Y = X_1\beta_1 + x_k\beta_k + \varepsilon$ → apply M_1 (FWL) → $\mathrm{Var}(b_k) = \sigma^2/(x_k^{*\prime}x_k^*) = \sigma^2/SST_{x_k^*}$ → $SST_{x_k^*}$ is the SSE of the auxiliary regression, which equals $(1-R_k^2)SST_{x_k}$.

(a) Formula (memorize):

$$\boxed{\mathrm{Var}(b_3 \mid X) = \frac{\sigma^2}{(1-R_3^2)\,SST_{x_3}}, \qquad SST_{x_3} = \sum_i (x_{i3}-\bar{x}_3)^2}$$

Here R_3^2 is the R^2 from regressing x_3 on *all the other regressors* (including the constant).

Where it comes from (L11.18–11.20), in case they ask for a derivation: Partition the model as $Y = X_1\beta_1 + x_k\beta_k + \varepsilon$. Multiply by M_1: $M_1Y = x_k^*\beta_k + M_1\varepsilon$, where $x_k^* = M_1x_k$ is the residual from regressing x_k on X_1. So $\mathrm{Var}(b_k \mid X) = \sigma^2(x_k^{*\prime}x_k^*)^{-1} = \sigma^2/SST_{x_k^*}$. Because X_1 has a constant,

x_k^* has mean 0, so $x_k^{*\prime}x_k^*$ is the SSE of that auxiliary regression. Since $R_k^2 = 1 - \frac{SSE_{aux}}{SST_{x_k}}$, we get $SSE_{aux} = (1 - R_k^2)SST_{x_k}$.

(b) The three components, and how each one improves precision:

1. σ^2, the variance of the unobserved part of y. Precision improves when σ^2 is smaller, for example by adding relevant controls that explain y and so take variation out of the disturbance.
2. $SST_{x_k} = (n-1)s_{x_k}^2$, the total observed variation in x_k. Precision improves with more variation in x_k or a larger sample.
3. $1 - R_k^2$, the share of the variation in x_k that is *not* explained by the other regressors. R_k^2 comes from regressing x_k on all the other x's and measures how well they predict x_k. OLS identifies β_k only from the *unique* variation in x_k, the part left after partialling out the others. Precision improves when R_k^2 is low, meaning x_k is not highly correlated with the other regressors. As $R_k^2 \to 1$, $\mathrm{Var}(b_k) \to \infty$. At $R_k^2 = 1$, A2 (full rank) fails. $\mathrm{VIF}_k = 1/(1 - R_k^2)$.

P14. Multicollinearity and the bias–variance trade-off

L11 Q5(a)–(b)

Skeleton

(a) Define multicollinearity as high $R_k^2 \to$ the denominator shrinks $\to$ the variance rises. (b) Dropping a correlated regressor lowers R_k^2 and the variance, but it moves that regressor into ε; since it is predictable from x_k, this breaks A3 and gives OVB, $\mathrm{E}(\Gamma)\beta_2$.

(a) Multicollinearity is the degree to which a regressor is (linearly) correlated with the other regressors, measured by R_k^2. From $\mathrm{Var}(b_k \mid X) = \sigma^2/[(1 - R_k^2)SST_{x_k}]$: as $R_k^2 \uparrow$, $(1 - R_k^2) \downarrow$, so $\mathrm{Var}(b_k) \uparrow$ and the estimates become less precise. Warning signs are a high overall R^2 with few significant coefficients, unstable estimates, and odd signs. The model is "close to" violating full rank. The effect can be offset by a smaller σ^2 or a larger SST_{x_k}.

(b) Dropping a highly correlated regressor x_j lowers R_k^2, which raises the denominator and *lowers* $\mathrm{Var}(b_k)$. But x_j now moves from the explained part of the model into the disturbance. Because x_j is highly predictable from x_k (that is what multicollinearity means), x_k now predicts the disturbance, so $\mathrm{E}(\varepsilon \mid X) \neq 0$. Mean independence (A3) fails, and b_k becomes biased. From the OVB formula (L10.12), the bias is $\mathrm{E}(b_1 - \beta_1) = \mathrm{E}(\Gamma)\beta_2$, which is nonzero exactly when the dropped variable matters ($\beta_2 \neq 0$) and is predicted by the kept ones ($\Gamma \neq 0$). This is the trade-off: lower variance in exchange for bias.

Lecture 12: OLS in Practice

P15. Log functional forms: log-level, level-log, log-log

L12 Q1–Q3

Skeleton

Exponentiate or differentiate $\to$ the chain rule gives the marginal effect $\to$ rearrange into "%-change" form, $100\,\partial y/y$ and/or $100\,\partial x/x \to$ read off the interpretation (watch the units).

Log-level, $\log(fare) = 4.666 + 0.043\,dist$ (dist in hundreds of miles)

(a) $fare = \exp(4.666 + 0.043\,dist)$.

(b) $\dfrac{\partial fare}{\partial dist} = \exp(4.666 + 0.043\,dist) \cdot 0.043 = 0.043\,fare$.

(c) Rearrange (b): $100\dfrac{\partial fare}{fare} = 100(0.043)\,\partial dist = 4.3\,\partial dist$. One unit of *dist* (100 miles) gives a $\approx 4.3\%$ increase. One mile is $\partial dist = 1/100 = 0.01$, so $100\dfrac{\partial fare}{fare} = 4.3 \times 0.01 = 0.043\%$. So the coefficient, read in percent, is the effect of *one mile*: $\approx 0.043\%$.

Level-log, $fare = 41 + 66\log(dist)$

(a) $\dfrac{\partial fare}{\partial dist} = \dfrac{66}{dist}$.

(b) Rearrange: $\dfrac{\partial fare}{\partial dist/dist} = 66$, so $\dfrac{\partial fare}{100\,\partial dist/dist} = \dfrac{66}{100} = 0.66$. A 1% increase in *dist* raises *fare* by $\approx$ \$0.66.

(c) Units don't matter. In miles, the hundreds-unit variable is $dist/100$: $fare = 41 + 66\log(dist/100) = \big(41 - 66\log 100\big) + 66\log(dist)$. Only the intercept changes. The slope is still 66, so a 1% increase still gives $\approx$ \$0.66.

Log-log, $\log(fare) = 4.25 + 0.40\log(dist)$ *(the parts with no posted solution)*

(a) $fare = \exp(4.25 + 0.40\log dist) = e^{4.25}\,dist^{0.40}$ ($\approx 70.1\,dist^{0.40}$).

(b) $\dfrac{\partial fare}{\partial dist} = 0.40\,e^{4.25} dist^{0.40-1} = 0.40\,\dfrac{e^{4.25} dist^{0.40}}{dist} = 0.40\,\dfrac{fare}{dist}$.

(c) Rearrange: $\dfrac{\partial fare/fare}{\partial dist/dist} = 0.40$, so $100\dfrac{\partial fare}{fare} = 0.40 \times 100\dfrac{\partial dist}{dist}$. This is an *elasticity*. Setting $100\,\partial dist/dist = 10$ gives $100\,\partial fare/fare = 0.40 \times 10 = 4$, so a 10% increase in distance means $\approx 4\%$ higher fare.

Model	Marginal effect	Interpretation of β	Units matter?
$y = \ldots + \beta x$	β	$+1$ unit $x \Rightarrow +\beta$ units y	yes
$\log y = \ldots + \beta x$	βy	$+1$ unit $x \Rightarrow \approx 100\beta\%$ change in y	yes (in x)
$y = \ldots + \beta \log x$	β/x	$+1\%$ in $x \Rightarrow \approx \beta/100$ units of y	no
$\log y = \ldots + \beta \log x$	$\beta y/x$	$+1\%$ in $x \Rightarrow \approx \beta\%$ change in y (elasticity)	no
LPM ($y \in \{0, 1\}$)	β	$+1$ unit $x \Rightarrow 100\beta$ *percentage points*	

Non-constant marginal effects: quadratic $\partial y/\partial x = \beta_2 + 2\beta_3 x$; interaction $\partial y/\partial x_k = \beta_2 + \beta_4 x_j$.

3 One-page formula sheet

OLS	$b = (X'X)^{-1}X'Y$; FOCs $X'e = 0$; $\hat{Y} = Xb = P_X Y$; $e = M_X Y = M_X \boldsymbol{\varepsilon}$
With a constant	$\sum e_i = 0$, $\bar{y} = \bar{x}'b$, $\widehat{\mathrm{Cov}}(x_k, e) = 0$
Bivariate	$b_2 = \dfrac{\overline{xy} - \bar{x}\bar{y}}{\overline{x^2} - \bar{x}^2} = \dfrac{\widehat{\mathrm{Cov}}(x,y)}{\widehat{\mathrm{Var}}(x)}$, $b_1 = \bar{y} - b_2\bar{x}$
Best predictors	$\arg\min_a \mathrm{E}(y-a)^2 = \mathrm{E}y$; $\arg\min_g \mathrm{E}(y-g(x))^2 = \mathrm{E}(y\|x)$
Sampling error	$b = \beta + (X'X)^{-1}X'\boldsymbol{\varepsilon} = \beta + (X'X/n)^{-1}\bar{w}$, $w_i = x_i\varepsilon_i$
Unbiased (A1–A3)	$\mathrm{E}(b\|X) = \beta \Rightarrow \mathrm{E}(b) = \beta$ (LIE)
Sandwich	$\mathrm{Var}(b\|X) = (X'X)^{-1}X'\mathrm{E}(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'\|X)X(X'X)^{-1}$
Classical (A4)	$\mathrm{E}(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'\|X) = \sigma^2 I_n \Rightarrow \mathrm{Var}(b\|X) = \sigma^2(X'X)^{-1}$
Robust (L13)	$\widehat{\mathrm{Var}}(b\|X) = \frac{n}{n-K}(X'X)^{-1}X'\mathrm{diag}(e_i^2)X(X'X)^{-1}$
Consistency / normality	LLN: $\mathrm{plim}\,\bar{w} = 0 \Rightarrow \mathrm{plim}\,b = \beta$; CLT: $b \overset{a}{\sim} N$
P and M	symmetric, idempotent; $P_X X = X$, $M_X X = 0$; $\mathrm{tr}(M_X) = n - K$
Constant-only P	$P_\iota = \frac{1}{n}\iota\iota'$ (all entries $1/n$); $P_\iota x = \bar{x}\iota$; $M_\iota x = x - \bar{x}$
s^2	$s^2 = \frac{e'e}{n-K}$, $\mathrm{E}(e'e\|X) = \sigma^2(n-K)$
Inference	$\frac{b_k - \beta_k}{\sqrt{\sigma^2[(X'X)^{-1}]_{kk}}} \sim N(0,1)$; $\frac{b_k-\beta_k}{\widehat{SE}} \sim t_{n-K}$
CI / test	$b_k \pm \kappa\widehat{SE}$; $t = \frac{b_k - q}{\widehat{SE}}$; $p = 2[1 - F_{n-K}(\|t\|)]$
Bivariate variance	$\mathrm{Var}(b_2\|X) = \sigma^2/\sum(x_i - \bar{x})^2 = \sigma^2/[(n-1)s_x^2]$
Multiple variance	$\mathrm{Var}(b_k\|X) = \dfrac{\sigma^2}{(1-R_k^2)SST_{x_k}}$, $\mathrm{VIF}_k = \frac{1}{1-R_k^2}$
FWL	$b_1 = (X_1^{*\prime}X_1^*)^{-1}X_1^{*\prime}Y^*$, $X_1^* = M_2X_1$, $Y^* = M_2Y$
OVB	$b_1 = \beta_1 + \Gamma\beta_2 + \delta$; $\mathrm{E}(b_1 - \beta_1) = \mathrm{E}(\Gamma)\beta_2$, $\Gamma = (X_1'X_1)^{-1}X_1'X_2$
Meas. error	$\mathrm{E}(b_2) = [1 - \mathrm{E}(\gamma_2)]\beta_2$, $0 < \mathrm{E}\gamma_2 < 1 \Rightarrow$ attenuation toward 0
Fit	$R^2 = 1 - \frac{SSE}{SST}$; adj-$R^2 = 1 - \frac{SSE/(n-K)}{SST/(n-1)}$

4 Skeleton drill (cover the right column)

Prompt	Chain of moves
P1 $\arg\min_a \mathrm{E}(y-a)^2$	$y=\mu+u$; expand; $\mathrm{E}u=0$ kills the cross term; $\mathrm{E}u^2+(\mu-a)^2$
P2 CEF best predictor	$\mathrm{E}(u\|x)=0$; $\mathrm{E}(hu)=\mathrm{E}_x[h\mathrm{E}(u\|x)]=0$; cross term with $h=m-g$
P3 residual facts	FOC $\sum x_{ik}e_i=0$; k=1: $\sum e=0$; $\bar{e}=\bar{y}-\bar{x}'b$; Cov expands to FOC $-\bar{x}_k\bar{e}$
P4 bivariate matrix	write out $X'X$, $X'Y$; factor n; 2×2 inverse; row 2
P5 unbiased	substitute model; pull out $f(X)$; A3; LIE
P6 variance	$b-\beta=A\boldsymbol{\varepsilon}$; $A\mathrm{E}(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}')A'$; $\mathrm{E}\boldsymbol{\varepsilon}=0$; $\sigma^2 I$; cancel
P7 LLN/CLT	$X'\boldsymbol{\varepsilon}=n\bar{w}$; $(X'X/n)^{-1}\bar{w}$; $\mathrm{E}w_i=0$ by LIE; plims; CLT
P8 $P_XH=0$	$H=Y-Xb$; $P_XY=Xb$; $P_XX=X$; $a=b-b$
P9 P_Z with ones	$Z'Z=n$; all entries $1/n$; $P_ZX=\bar{x}$; $X^*=x_i-\bar{x}$
P10 standardize	$N(0,1)$ with σ^2; t_{n-K} with s^2
P11 s^2 unbiased	$M'M=M$ (expand); trace, cycle, E; $\sigma^2\mathrm{tr}M$; $n-K$
P12 bivariate Var	$(X'X)^{-1}$ (2,2) entry; det $=n\,SST_x$; three factors
P13 $\mathrm{Var}(b_k)$	FWL $\to \sigma^2/SST_{x^*}$; $SSE_{aux}=(1-R_k^2)SST$; three parts
P14 multicollinearity	$R_k^2\uparrow\Rightarrow$ Var $\uparrow$; drop $\Rightarrow$ A3 fails $\Rightarrow$ OVB
P15 logs	exponentiate; chain rule; $100\partial y/y$ vs $100\partial x/x$; units

5 Mock proof exam (permuted, closed book)

Give yourself about 50 minutes. Write every step and cite the assumption you use at each one.

1. Let $m(x)=\mathrm{E}(y\mid x)$ and $u=y-m(x)$. Show that $\mathrm{E}(u\mid x)=0$. Then show that no function $g(x)$ predicts y with smaller expected squared error than $m(x)$. *(P2)*

2. Starting from $Y=X\beta+\boldsymbol{\varepsilon}$ and $b=(X'X)^{-1}X'Y$: (a) show $b=\beta+(X'X)^{-1}X'\boldsymbol{\varepsilon}$; (b) show $\mathrm{E}(b)=\beta$, stating where you use mean independence and the LIE; (c) derive $\mathrm{Var}(b\mid X)=\sigma^2(X'X)^{-1}$ under homoskedasticity and no autocorrelation. *(P5 + P6)*

3. Show $e=M_X\boldsymbol{\varepsilon}$, then $e'e=\boldsymbol{\varepsilon}'M_X\boldsymbol{\varepsilon}$, then $\mathrm{E}\left(\frac{e'e}{n-K}\mid X\right)=\sigma^2$. Use no shortcuts. Finally, explain why replacing σ^2 with s^2 in the standardized estimate changes $N(0,1)$ to t_{n-K}. *(P11 + P10)*

4. In $y_i=\beta_1+\beta_2x_i+\varepsilon_i$: (a) derive $b_2=\widehat{\mathrm{Cov}}(x,y)/\widehat{\mathrm{Var}}(x)$ from $(X'X)^{-1}X'Y$; (b) derive $\mathrm{Var}(b_2\mid X)=\sigma^2/\sum(x_i-\bar{x})^2$ and name the three factors. *(P4 + P12)*

5. Let Z be a column of ones. Describe P_Z, P_ZX, and M_ZX completely. Use the result to explain how OLS "holds other variables fixed." Then write the formula for $\mathrm{Var}(b_k\mid X)$ in a multiple regression and explain what R_k^2 represents. *(P9 + P13)*

Bonus quick hits: (i) Show $P_X H = 0$ and that regressing e on X gives $a = 0$. (ii) In $\log(fare) = 4.25 + 0.40\log(dist)$, show that a 10% increase in distance gives about a 4% higher fare. (iii) Explain why dropping a collinear regressor can introduce bias.

Source files: `Exercises/Short_Answers/PDF/` (Canvas SAQ questions and solutions, Lectures 08–12) and `Lecture_Notes/lectureECON5021.pdf` (Oct 14, 2026), Lectures 8–13.